

Properties of Rhodium(II) Complexes Having Cytostatic Activity

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ABSTRACT

Dimeric rhodium(II) carboxylate complexes $[\text{Rh}_2(\text{OOCR})_4\text{L}_2]$, $[\text{Rh}_2(\text{OOCR})_2(\text{N-N})_2\text{L}_2]^{2+}$ and $[\text{Rh}_2\text{Cl}_2(\text{RCOO})_2(\text{N-N})_2]$ ($\text{R} = \text{PhCH}(\text{OH})$, $\text{MeCH}(\text{OH})$, $\text{HOOC-CH}(\text{OH})\text{CH}(\text{OH})$, Me ; $\text{N-N} = 2,2'$ -bipyridine, 1,10-phenanthroline) have been synthesized and characterized using electronic, NMR, and IR spectra. Cytostatic activity of these complexes has been tested for human oral carcinoma KB cell line. Some complexes with hydroxycarboxylates and $[\text{Rh}_2(\text{RCOO})_2(\text{N-N})_2]^{2+}$ compounds show higher cytostatic activity than $[\text{Rh}_2(\text{OOCCH}_3)_4(\text{H}_2\text{O})_2]$.

INTRODUCTION

Rhodium(II) carboxylato complexes such as $[\text{Rh}_2(\text{OOCR})_4\text{L}_2]$ [1], $[\text{Rh}_2\text{X}_2(\text{OOCR})_2(\text{N-N})_2]$, [2], and $[\text{Rh}_2(\text{OOCR})_2(\text{L-L})_2]$ [1, 3] where $\text{L} = \text{Lewis base}$, $\text{X} = \text{halide}$, $\text{N-N} = 2,2'$ -bipyridine (bpy), 1,10-phenanthroline (phen), $\text{L-L} = \text{acetylacetonate}$, dimethylglyoximate, etc., have aroused considerable interest because of their interesting structure [1–3] and catalytic properties [1, 4].

The $\text{Rh}_2(\text{OOCR})_4$ complexes exhibited anticancer activity [1, 5–8]. The activity was found to be in order: methoxyacetate < propionate < butyrate [5]. This correlates with the thermodynamic stabilities of the complexes with adenosine 5-phosphates and imidazole [1, 5]. A close correlation has also been observed between the deactivation of a sulfhydryl enzyme and antitumor activity of rhodium(II) carboxylates [5]. However, it was suggested that formation of hydrogen bonds between nucleobases and oxygen atoms of carboxylato ligands is responsible for antitumor activity as well [9]. We have prepared and investigated structure, reactivity, and catalytic activity of $[\text{Rh}_2\text{X}_2(\text{OOCR})_2(\text{N-N})_2]$ ($\text{R} = \text{H}$, Me , PhCHOH , $\text{N-N} = 2,2'$ -bipyridine, 1,10-phenanthroline, and their derivatives) [2, 4]. The purpose of the present work is to synthesize and to study properties of cationic complexes $[\text{Rh}_2(\text{OOCR})_2(\text{N-N})_2(\text{H}_2\text{O})_2]^{2+}$ and investi-

gate the cytostatic activity of these complexes as well as compounds $[\text{Rh}_2\text{X}_2(\text{OOCR})_2(\text{N-N})_2]$ containing Cl, Br, or I ligands coordinated along the Rh-Rh bond.

EXPERIMENTAL

Complexes $\text{Rh}_2(\text{OAc})_4 \cdot 2\text{H}_2\text{O}$ [10], $\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4 \cdot 2.5\text{H}_2\text{O}]$ [11], $[\text{Rh}_2(\text{OOCR})_4 \cdot 2\text{H}_2\text{O}]$ [2a], $[\text{Rh}_2\text{Cl}_2(\text{OOCR})_2(\text{bpy})_2] \cdot 2\text{H}_2\text{O}$, and $[\text{Rh}_2\text{Cl}_2(\text{OOCR})_2(\text{phen})_2] \cdot 4\text{H}_2\text{O}$ ($\text{R} = (S)\text{-PhCH}(\text{OH})$, H [2] were prepared by the literature methods. Rhodium(II) lactate, tartrate, malate, and citrate were prepared in situ in water solutions in reactions $\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4]$ with the appropriate acids: $(S)\text{-MeCH}(\text{OH})\text{COOH}$, $(2R, 3R)\text{-(+)-HOOCCH}(\text{OH})\text{CH}(\text{OH})\text{COOH}$, $(S)\text{-HOOCCH}(\text{OH})\text{CH}_2\text{COOH}$, and $\text{HOOCCH}_2\text{C}(\text{OH})(\text{COOH})\text{CH}_2\text{COOH}$.

Rhodium(II) formate $\text{Rh}_2(\text{HCOO})_4 \cdot 2\text{H}_2\text{O}$ was prepared in reaction of $\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4] \cdot 2.5\text{H}_2\text{O}$ (1 mmol, 0.583 g) with 8.5 cm³ formic acid (6%). Product precipitated as a green solid and was filtered off, washed with water, and then with methanol. Yield: 0.315 g (75%).

Complexes $[\text{Rh}_2(\text{OOCMe})_2(\text{N-N})_2\text{X}_2]$ ($\text{N-N} = 2,2'\text{-bipyridine (bpy)}$, $1,10\text{-phenanthroline (phen)}$; $\text{X} = \text{Cl, Br, I}$) have been obtained in the reaction of $\text{Rh}_2(\text{OAc})_4 \cdot 2\text{H}_2\text{O}$ (10^{-4} mol) and NaX ($2.2 \cdot 10^{-4}$ mol) in 75% EtOH (10 cm³). Mixtures were refluxed under argon for 3 hr and filtered in air atmosphere. The resulting reddish-brown solutions gave red-brown or brown products. They were washed with water and methanol; yield: 60–80%.

$[\text{Rh}_2\{\text{OOCCH}(\text{OH})\text{C}_6\text{H}_5\}_2(\text{phen})_2(\text{H}_2\text{O})_2]\{\text{OOCH}(\text{OH})\text{C}_6\text{H}_5\}_2$. Solution of $[\text{Rh}_2(\text{O}_2\text{CCH}(\text{OH})\text{C}_6\text{H}_5)_4(\text{H}_2\text{O})_2]$ (4.23 g, $5 \cdot 10^{-3}$ mol) and $1,10\text{-phenanthroline}$ (1.98 g, 10^{-2} mol) in 100 cm³ of absolute ethanol was heated at reflux for 3 hr, cooled to room temperature, and filtered. From the filtrate, the yellow-brown product was deposited by means of diethyl ether. Precipitate was filtered off, washed with $(\text{C}_2\text{H}_5)_2\text{O}$, and dried in air; yield: 5.42 g (90%).

Infrared spectra (KBr pellets) were measured on Perkin Elmer 283 and Carl Zeiss M 80 instruments, UV-VIS on Cary 5, and Beckman UV 5240 spectrophotometers.

INVESTIGATION OF CYTOSTATIC ACTIVITY

Cells and Culture Conditions

Human KB cell line (oral cavity carcinoma cells) was maintained as a monolayer culture in MEM Eagle's medium supplemented with 10% of fetal bovine serum (FBS). Cells were passaged with the use of trypsin-EDTA mixture.

Inhibition of Protein Synthesis Assay

KB cell cultures ($4 \cdot 10^6$ cells/2 cm³ of growth medium) were started in Leighton tubes. After 3 hr, the tested compounds were added in the maximal solvent volume of 0.1 cm³. To the control tubes an appropriate volume of the solvent alone was added. The tubes were incubated in 5% CO₂/95% air atmosphere at 37° for 72 hr. Initially, the compounds were tested at 100, 10, and 1 μg/cm³

concentrations. For the second and further tests, compounds were tested at five doses at twofold dilutions [13].

The growth of cells in the culture was measured by estimation of total cellular protein, according to the modified Lowry method [14] with bovine serum albumin (Serva, F.R.G.) as a standard. Percentage of the growth inhibition was estimated according to the equation [12, 13].

$$\% \text{ of inhibition of protein synthesis} = \frac{(C_n - T_n)}{C_n - T_0} 100$$

where C_n is the amount of protein in the control culture at the end of the experiment (after time n), T_0 is the amount of protein in the culture at the beginning of the experiment (at time 0), and T_n is the amount of protein in the test culture at the end of the experiment (after time n).

As a regular control group, medium blank tubes and untreated cultures were analyzed. Test tubes were read when in the controls (after time n), the protein content was at least six times higher than in baseline tubes (after time 0).

A compound concentration causing 50% inhibition of the protein growth (ED_{50} , effective dose 50%) was estimated from the percent of inhibition of the protein synthesis by the graphic method, with the use of probits table [15, 16].

RESULTS AND DISCUSSION

The metal-metal bonded tetra- μ -carboxylato-dirhodium and $[Rh_2(OOCR)_2(N-N)_2X_2]$ complexes (Figure 1) readily form adducts in which the axial sites are occupied by donor solvents or ligands. Substitution of axial ligands proceeds readily (Figure 1) and reversibly.

The UV-VIS spectra of rhodium(II) carboxylates are very characteristic. There are two bands in the visible region (Table 1). The energy of band I (usually observed in the region $1.5\text{--}1.9 \mu\text{m}^{-1}$) strongly depends on the nature of axial ligands, while the energy of band II ($2.2\text{--}2.4 \mu\text{m}^{-1}$) is insensitive to the properties of these ligands. The energy of band I has been shown previously to increase with increasing field strength of axial ligand [1, 2]: $O < N < P$. Therefore, the exchange of axial ligands may be easily observed using electronic spectroscopy. Substitution of the bridging ligands is more difficult. The $Rh_2(OOCR)_2(N-N)_2X_2$ species (Figure 1) are formed after relatively long

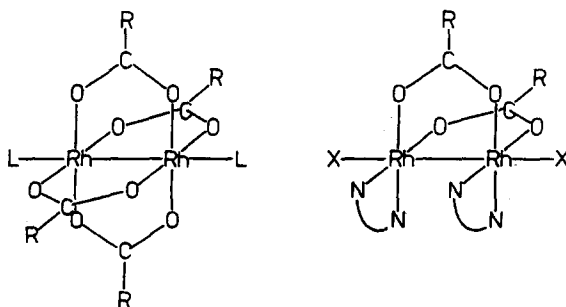


FIGURE 1. Structure of $Rh_2(OOCR)_4L_2$ and $Rh_2(OOCR)_2(N-N)_2X_2$.

TABLE 1. Electronic Spectra of Rhodium(II) Complexes

Complex/Solvent	Bands μm^{-1} , $\epsilon(10^{-2} \text{ M}^{-1} \cdot \text{cm}^{-1})$
$\text{Rh}_2(\text{OOCCHOHPh})_4/\text{EtOH}$	1.70(2.60), 2.22(1.20), 2.44(0.80), 3.00(1.70), 3.80(43), 3.89(51), 3.98(52), 4.31(103)
$\text{Rh}_2(\text{OOCCH}_3)_2(\text{phen})_2\text{Cl}_2/\text{H}_2\text{O}$	1.76(2.60), 2.45(27.5), 2.76(45), 3.71(350), 3.97(466), 4.44(466)
$\text{Rh}_2(\text{OOCCHOHPh})_2(\text{bpy})_2\text{Cl}_2/\text{EtOH}$	1.80(4.00), 2.30(30), 2.65(53), 3.23(230), 3.66(394)
$\text{Rh}_2(\text{OOCCHOHPh})_2(\text{bpy})_2\text{Br}_2/\text{MeOH}$	1.80(3.30), 2.33(33), 2.60(55), 3.03(166), 3.24(219), 3.66(320)
$\text{Rh}_2(\text{OOCCHOHPh})_2(\text{bpy})_2\text{I}_2/\text{MeOH}$	1.83(7.50), 2.21(67), 2.50(144), 2.79(134), 3.25(126), 3.58(230), 3.79(207), 3.86(198)
$\text{Rh}_2(\text{OOCCHOHPh})_2(\text{phen})_2\text{Cl}_2/\text{EtOH}$	1.80(3.70), 2.30(32), 2.63(54), 3.29(188), 3.55(275), 3.68(319), 3.86(396)
$\text{Rh}_2(\text{OOCCHOHPh})_2(\text{phen})_2\text{Br}_2/\text{MeOH}$	1.80(2.00), 2.31(25), 2.62(42), 3.15(117), 3.79(192), 3.89(265)
$\text{Rh}_2(\text{OOCCHOHPh})_2(\text{phen})_2\text{I}_2/\text{MeOH}$	1.81(4.20), 2.22(50), 2.55(92), 2.80(97), 3.31(102), 3.88(278)
$[\text{Rh}_2(\text{OOCCHOHPh})_2(\text{phen})_2(\text{H}_2\text{O})_2]\text{-(PhCHOHCOO)}_2/\text{EtOH}$	1.77(2.60), 2.30(22), 2.82(56), 3.77(400)
$\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4] \cdot 2.5\text{H}_2\text{O} + 4\text{MeCHOHCOOH}$	1.70(2.10), 2.22(1.08), 3.63(23.5), 3.96(40.0)
$\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4] \cdot 2.5\text{H}_2\text{O} + 4\text{MeCHOHCOOH} + 2\text{phen}$	1.80(2.80), 2.30(24), 2.70, 3.00, 3.38, 3.72
$[\text{Rh}_2(\text{OOCCH}_3)_2(\text{phen})_2(\text{H}_2\text{O})_2](\text{CH}_3\text{COOH})$	1.80(3.30), 2.30(26.0), 2.68, 2.95, 3.40, 3.70, 3.87(455)

heating of $[\text{Rh}_2(\text{OOCR})_4\text{L}_2]$ compounds with N-N and X ligands. The band I assigned to the $\pi^*(\text{Rh}_2) \rightarrow \sigma^*(\text{Rh}_2)$ transition is observed at slightly higher energy (ca. $0.1 \mu\text{m}^{-1}$) in the spectra of $\text{Rh}_2(\text{OOCR})_2(\text{N-N})_2\text{X}_2$ complexes than in the spectra of $\text{Rh}_2(\text{OOCR})_4$ compounds since $\pi^*(\text{Rh}_2)$ orbitals are stabilized because of interaction with π^* orbitals of nitrogen ligands [17]. However, the position of band I in the spectra of $[\text{Rh}_2\text{X}_2(\text{OOCR})_2(\text{N-N})_2]$ does not change essentially for a particular halo (X) ligand. This indicates a weak interaction of axial halo ligands with rhodium atoms. This was confirmed by the Fenske-Hall calculation [17]. At about $2.3 \mu\text{m}^{-1}$, an intensive band is observed, which could be assigned to the $\sigma(\text{Rh}_2) \rightarrow \pi^*(\text{N}=\text{C})$ transition, i.e., to the electron transfer from rhodium to the empty π^* orbitals of the nitrogen ligands (bpy or phen). The ligand-to-metal charge transfer bands $\sigma(\text{Rh-X}) \rightarrow \sigma^*(\text{Rh}_2)$ for Cl, Br, and I complexes are observed at about 3.7, 3.2, and $2.5 \mu\text{m}^{-1}$ in H_2O and MeOH solutions. This indicates that substitution of axial halide ligands is more difficult than oxygen ligands.

Solutions of chloro and bromo complexes give conductivities in the $140 \Omega^{-1} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$ range, while the iodo species gave about half this value.

All the halide compounds show incomplete dissociation. The prevailing species in solution are $[\text{Rh}_2(\text{X})(\text{OOCR})_2(\text{N-N})_2(\text{H}_2\text{O})_2]^+$ cationic complexes.

The IR spectra showed that carboxylate ligands are symmetrically bonded to both rhodium atoms because differences between $\nu^{\text{as}}(\text{COO})$ and $\nu^{\text{s}}(\text{COO})$ are small both in complexes with halides and in cationic complexes.

The structure of $[\text{Rh}_2(\text{OOCR})_2(\text{N-N})_2(\text{H}_2\text{O})_2]^{2+}(\text{RCOO}^-)_2$ has been confirmed not only by electronic spectra, but also by IR (Table 2) and ^1H NMR spectra, which showed the presence of noncoordinated RCOO^- anions.

TABLE 2. IR Spectra of Rh(II) Compounds

Compound	$\nu(\text{CH})$	$\nu^{\text{as}}(\text{COO})$	$\nu^{\text{s}}(\text{COO})$	$\nu(\text{Rh-halide})$
$[\text{Rh}_2(\text{OOCCHOHPh})_2(\text{bpy})_2\text{Cl}_2] \cdot 2\text{H}_2\text{O}$	3090, 3050, 3020, 2915	1570	1405	195
$[\text{Rh}_2(\text{OOCCHOHPh})_2(\text{bpy})_2\text{Br}_2] \cdot 2\text{H}_2\text{O}$	3065, 3030, 2920	1580	1405	126, 132
$[\text{Rh}_2(\text{OOCCHOHPh})_2(\text{bpy})_2\text{I}_2] \cdot 2\text{H}_2\text{O}$	3070, 3030, 2920	1575	1405	109
$[\text{Rh}_2(\text{OOCCHOHPh})_2(\text{phen})_2\text{Cl}_2] \cdot 4\text{H}_2\text{O}$	3070, 3045, 3015, 2920	1575	1410	188
$[\text{Rh}_2(\text{OOCCHOHPh})_2(\text{phen})_2\text{Br}_2] \cdot 4\text{H}_2\text{O}$	3070, 3050, 3020, 2920	1575	1410	133
$[\text{Rh}_2(\text{OOCCHOHPh})_2(\text{phen})_2\text{I}_2] \cdot \text{H}_2\text{O}$	3075, 3050, 3020, 2920	1570	1410	104
$[\text{Rh}_2(\text{OOCCHOHPh})_2(\text{phen})_2(\text{H}_2\text{O})_2] \cdot (\text{PhCHOHCOO})_2$	3085, 3060, 3020, 2975, 2920	1607vs, 1592vs, 1566vs	1420vs, 1400vs, 1372vs	
$[\text{Rh}_2(\text{OOCCH}_3)_2(\text{phen})_2(\text{H}_2\text{O})_2] \cdot (\text{CH}_3\text{COO})_2$	3090, 3063, 3020, 2935	1597s, 1585s, 1568vs, 1560vs	1433vs, 1412vs	

In ^1H NMR spectra, chemical shifts of CH_3 and $\text{PhCH}(\text{OH})$ protons are higher for 0.8–1.0 ppm than those for noncoordinated carboxylates: $[\text{Rh}_2(\text{PhCHOHCOO})_2(\text{phen})_2(\text{H}_2\text{O})_2][\text{PhCHOHCOO}]_2$ {4.7s(CH), 5.70s(CH), 6.55–9.5m($\text{C}_6\text{H}_5 + \text{phen}$)}; $[\text{Rh}_2(\text{CH}_3\text{COO})_2(\text{phen})_2(\text{H}_2\text{O})_2][\text{CH}_3\text{COO}]_2$ {1.80s(CH_3), 2.65s(CH_3), 7.67dd(H^3 , $J_{2,3} = 5.0$ Hz, $J_{3,4} = 9.0$ Hz, 7.75s(H^5), 8.33dd(H^4 , $J_{2,4} = 1$ Hz), 8.61d(H^2)}

Rhodium(II) carboxylates $[\text{Rh}_2(\text{OOCR})_4]$ exhibit antitumor activity [1, 5]. Hitherto activity of complexes with aliphatic carboxylates was investigated. We have found that cytostatic activity of complexes with hydroxycarboxylates and $[\text{Rh}_2(\text{RCOO})_2(\text{N-N})_2(\text{H}_2\text{O})_2]^{2+}$ compounds against KB cell line in vitro in some cases is not lower than that of $\text{Rh}_2(\text{OOCMe})_4$. The obtained results indicate that rhodium complexes and α -hydroxycarboxylic acids ($\text{PhCH}(\text{OH})\text{COOH}$ and $\text{MeCH}(\text{OH})\text{COOH}$) show higher cytostatic activity than complexes with other hydroxyacids containing two or three carboxylate groups (Table 3). Substitution of two carboxylate bridging ligands with phen enhances cytostatic activity of complexes $[\text{Rh}_2(\text{OOOCCH}(\text{OH})\text{R})_2(\text{phen})_2(\text{H}_2\text{O})_2]^{2+}$ ($\text{R} = \text{Me}, \text{Ph}$). However, the cytostatic activity of analogous bpy complexes is lower. Activity of $\text{Rh}_2(\text{OOCR})_2(\text{N-N})_2\text{X}_2$ complexes is low because X^- ligands are not completely split off. Electronic spectra of $[\text{Rh}_2(\text{OOCH}(\text{OH})\text{Ph})_2(\text{N-N})_2\text{Cl}_2]$ in the presence of adenine or adenosine ([complex]: [adenine] = 1:2) are identical with the spectra of complexes. This indicates that nucleobases added to the solution of complexes are not coordinated with rhodium atoms. Other carboxylate complexes coordinate adenine and adenosine molecules. It was evidenced by the blue shift of band I for about 40–50 nm for complexes with nucleobases as compared with water adducts $\text{Rh}_2(\text{OOCR})_4(\text{H}_2\text{O})_2$.

These data explain the low cytostatic activity of $\text{Rh}_2(\text{OOCR})_2(\text{N-N})_2\text{Cl}_2$ and relatively high activity of water adducts of rhodium(II) complexes. Higher

TABLE 3. Cytostatic Activity of New Dirhodium Derivatives for Human Oral Carcinoma KB Cell Line. Concentrations of Solutions Used in Experiment: [complex] = 10^{-3} mol · dm $^{-3}$; [ligand] = $2 \cdot 10^{-3}$ mol · dm $^{-3}$

Substance Tested		Ligand	ED $_{50} \cdot 10^6$
Rh $_2$ (CH $_3$ COO) $_4 \cdot 2$ H $_2$ O			30
Rh $_2$ (CH $_3$ COO) $_4 \cdot 2$ H $_2$ O	+	2,2'-bpy	60
Rh $_2$ (CH $_3$ COO) $_4 \cdot 2$ H $_2$ O	+	1,10-phen	85
Rh $_2$ (PhCHOHCOO) $_4 \cdot 2$ H $_2$ O	+	2,2'-bpy	> 100
Rh $_2$ (PhCHOHCOO) $_4 \cdot 2$ H $_2$ O	+	1,10-phen	8
Na $_4$ Rh $_2$ (CO $_3$) $_4 \cdot 2.5$ H $_2$ O	+	lactic acid*	15
Na $_4$ Rh $_2$ (CO $_3$) $_4 \cdot 2.5$ H $_2$ O	+	lactic acid* + 2,2'-bpy	53
Na $_4$ Rh $_2$ (CO $_3$) $_4 \cdot 2.5$ H $_2$ O	+	lactic acid* + 1,10-phen	8
Na $_4$ Rh $_2$ (CO $_3$) $_4 \cdot 2.5$ H $_2$ O	+	tartaric acid	55
Na $_4$ Rh $_2$ (CO $_3$) $_4 \cdot 2.5$ H $_2$ O	+	citric acid	> 100
Na $_4$ Rh $_2$ (CO $_3$) $_4 \cdot 2.5$ H $_2$ O	+	malic acid	> 100
Rh $_2$ (HCOO) $_4 \cdot 2$ H $_2$ O	+	2,2'-bpy	> 100
Rh $_2$ (HCOO) $_2$ (bpy) $_2$ Cl $_2$			> 100
Rh $_2$ (HCOO) $_2$ (phen) $_2$ Cl $_2$			> 100

* [MeCHOHCOOH] = $4 \cdot 10^{-3}$ mol · dm $^{-3}$.

cytostatic activity of complexes containing 1,0-phenanthroline and hydroxycarboxylate ligands, as compared with acetato-phenanthroline coordination compounds, seems to indicate, as suggested by Farrell [9], the important role played by the formation of hydrogen bonds between complexes and nucleobases.

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